P-C-447C
June 11, 1975
SUPERSEDING
Fed. Spec. P-C-447B
November 15, 1963

FEDERAL SPECIFICATION

CLEANING COMPOUND, TOILET BOWL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification describes two different types of toilet bowl cleaners (see 6.1).

1.2 Classification.

1.2.1 Types. The toilet bowl cleaners covered by this specification shall be of the following types, as specified:

Type I - Liquid.
Type II - Solid (globular or granular form).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 536/GEN - Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.
Fed. Test Method 536/2101 - Sieve Tests.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.
40 CFR 162 - Federal Insecticide, Fungicide, and Rodenticide Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I toilet bowl cleaner shall be a free flowing liquid. The liquid shall consist of a minimum of 13 percent and a maximum of 25 percent titratable hydrogen chloride. In addition, the liquid shall contain wetting agents and cleaning agents for the removal of organic soils from vitreous china.

3.1.2 Type II toilet bowl cleaner shall consist of technical grade sodium bisulfate.

3.2 Shelf life. For types I and II toilet bowl cleaners, the product and the container shall be completely stable and shall not deteriorate for a minimum period of one year when stored at 95°F.

3.3 Corrosion inhibition. The type I cleaning compound shall be correctly inhibited to minimize metallic corrosion. When the cleaner is tested as specified in the test procedure described in 4.4.1, the weight loss of the metal shall not exceed 0.50 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

3.4 Registration. The toilet bowl cleaners covered by this specification shall be labeled in compliance with the Federal Hazardous Substances Act, except that if the claim is made on the label that the toilet bowl cleaner can act as a disinfectant or sanitizer, the toilet bowl cleaner shall be registered with the Environmental Protection Agency under the Federal Insecticide, Fungicide, and Rodenticide Act before shipping.

3.5 Acidity (HCl). Type I cleaning compound shall contain a minimum of 13.0 percent and a maximum of 25.0 percent hydrogen chloride when tested as specified in 4.4.2.

3.6 Acidity (H_2SO_4). Type II cleaning compound shall contain a minimum of 30.0 percent sulfuric acid when tested as specified in 4.4.3.

3.7 Particle size. Type II cleaning compound shall conform to the particle size requirements in table I when tested as specified in 4.4.4.

TABLE I. Particle size.

Sieve number	Globular		Granular	
	Percent retained on		Percent retain on	
	Minimum	Maximum	Minimum	Maximum
8	---	---	---	5.0
10	0.0	5.0	---	---
20	---	---	30.0	60.0
40	90.0	---	60.0	---

3.8 Moisture content. The moisture content of the type II cleaning compound shall be a maximum of 1.50 percent when tested as specified in 4.4.5.

3.9 Iron content. The iron content of the type II cleaning compound shall be of a maximum of 0.15 percent when tested as specified in 4.4.6.

3.10 Insoluble matter. The type II cleaning compound shall not contain more than 2.0 percent insoluble matter when tested as specified in 4.4.7.

3.11 Labeling. In addition to the requirements of 3.4, the following statement, or a statement equivalent in content shall be durably and legibly printed on the product label: Do not use on bathtubs, wash stands, sinks, or other metal or enameled fixtures.

3.12 Workmanship. The finished product shall conform to the levels of quality established herein. The occurrence of defects shall not exceed the acceptable quality levels (AQLs) established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sample for inspection and test shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling and testing, a lot shall consist of all material manufactured using the same type raw materials and the same manufacturing process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

4.3 Inspection.

4.3.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 Testing and inspection of the end item.

4.3.2.1 Testing of the end item. The end item shall be tested in accordance with the test methods specified in table II. When four or more containers compose the shipping lot, a sample of not less than one pound shall be taken from each of not less than four separate shipping containers, randomly selected throughout the lot presented for examination. When less than four shipping containers comprise the lot the total sample shall, in all cases, consist of not less than three sample portions taken at random from the container or containers and shall be placed in a suitable size receptacle. After thorough mixing, a two pound composite sample shall be drawn for testing purposes. The composite sample shall be placed in an air-tight container, sealed, and marked with the identifying lot number and date.

4.3.2.2 Inspection of the end item. The end item shall be examined for appearance in accordance with the classification of defects and acceptable quality levels (AQLs) set forth below. The lot size for determining the sample size shall be expressed in units of filled containers of the same capacity consisting of toilet bowl cleaner manufactured in one batch and offered for delivery at one time for the examination in 4.3.2.2.1. The inspection levels for purposes of determining the sample in accordance with the provisions of MIL-STD-105 and the acceptable quality levels (AQLs) shall be as specified in 4.3.2.2.2.

TABLE II. Instructions for testing

Characteristic	Specification reference		Requirements applicable to		Number of determinations per sample unit	Results reported as
	Requirement	Test Method	Ind. unit	Composite sample		Pass or Fail
Corrosion test	3.3	4.4.1		X	Avg. of 2	X
Acidity (HCl)	3.5	4.4.2		X	Avg. of 2	X
Acidity (H_2SO_4)	3.6	4.4.3		X	Avg. of 2	X
Particle size	3.7	4.4.4		X	Avg. of 2	X
Moisture content	3.8	4.4.5		X	Avg. of 2	X
Iron content	3.9	4.4.6		X	Avg. of 2	X
Insoluble matter	3.10	4.4.7		X	Avg. of 2	X

4.3.2.2.1 Examination of the end item for defects in appearance and odor. The sample unit for this examination shall be the contents of one filled container. Examine for defects listed below.

Examine	Defect
Appearance	In the case of a liquid toilet cleaner: Not a free flowing liquid; not of consistent color or viscosity. Presence of dirt of foreign material. In the case of a solid toilet bowl cleaner: Not free flowing; not uniform, lumpy or caked.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

4.3.2.2.2 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2.2.1	S-2	2.5

4.3.2.2.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0.

TABLE III. Examination for preparation for delivery.

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.4 Test methods.

4.4.1 Corrosion test. A brass coupon 1/ is cleaned with a 10 percent (by volume) hydrochloric acid solution, rinsed thoroughly with tap water and distilled water. A cast iron coupon 1/, is scrubbed with pumice using a brush, rinsed thoroughly with tap water and distilled water. The metals are placed in an oven at 105° for 30 minutes, cooled in a desiccator and weighed. The metals are immersed in the test solution 2/ for 24 hours, rinsed thoroughly with tap water and distilled water, dried in an oven at 105°C for 30 minutes, cooled in a desiccator and weighed. The degree of corrosion is measured by the percent loss in weight.

1/ The dimensions of the brass coupon are 2 in. x 1 in. x 1/16 in. The dimensions of the cast iron coupon are 2 in. x 1 in. x 1/8 in. The coupons can be obtained from the Chemical Specialties Manufacturer's Association, Inc., 1001 Connecticut Avenue, N.W. Washington, DC 20036.

2/ The test solution used in this test is the same as the test solution used in 4.4.1.

4.4.1.1 Calculation. Percent loss in weight = $\frac{\text{Weight loss}}{\text{Original weight}} \times 100$

4.4.2 Acidity (HCl). Weigh a glass stoppered Erlenmeyer flask containing about 30 ml of distilled water. Add rapidly 3 to 4 ml of the thoroughly mixed sample, stopper and reweigh. Dilute to 50 ml with distilled water and titrate with 1 N sodium hydroxide using methyl orange indicator.

4.4.2.1 Calculation.

$$\text{Percent HCl} = \frac{A \times N \times 0.03646 \times 100}{B}$$

where: A = Milliliters of sodium hydroxide solution.

N = Normality of sodium hydroxide solution.

B = Weight, in grams, of the specimen.

4.4.3 Acidity (H_2SO_4). Weigh quickly 10 g to the nearest mg of the sample, dissolve and dilute to 500 ml in a volumetric flask with distilled water, transfer a 50 ml aliquot portion to a 400 ml beaker, add about 200 ml of distilled water, and add 2 drops of a one percent phenolphthalein indicator solution. Titrate to a distinct pink end point with a 0.2N sodium hydroxide solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

4.4.3.1 Calculation. Percent H_2SO_4 = $\frac{T \times N \times 4.9}{W}$

where: T = ml of NaOH solution required for the titration.

N = Normality of NaOH solution.

W = Sample weight in aliquot.

4.4.4 Particle size. Sieving tests shall be determined in accordance with applicable provisions of method 2101 of Fed. Test Method Std. No. 536.

4.4.5 Moisture content. Place approximately 15 grams of the sample in a tared covered Petri dish and weigh on an analytical balance. Set the Petri dish in its cover and dry in an electric oven at 105°C + 5°C for 16 hours. Cool in a desiccator and reweigh.

4.4.5.1 Calculation. Percent H_2O = $\frac{\text{Loss in weight}}{\text{Weight of sample}} \times 100$

4.4.6 Iron content. Dissolve 5 grams of the sample in 60 ml of water and 20 ml of hydrochloric acid and boil gently for 10 minutes. Cool and dilute to 100 ml. Dilute 10 ml of this solution to 50 ml and add 30 to 50 mg of ammonium persulfate crystals and 3 ml of ammonium thiocyanate reagent solution. The ammonium thiocyanate reagent solution is made by dissolving 60 grams reagent grade ammonium thiocyanate in 1 liter of distilled water. Any red color shall not exceed that produced by 7.5 mg of iron subjected to the same analytical procedure.

4.4.7 Insoluble matter. Accurately weigh 50 grams of the sample and dissolve in 300 ml of hot distilled water in a 600 ml beaker. Allow the insoluble matter to settle, and filter by decanting through a tared Gooch crucible. Wash the insoluble matter into the crucible with more hot water. When the insoluble matter is completely filtered, place the crucible in an electric oven at 100-110°C for one hour. Cool in a desiccator and reweigh.

4.4.7.1 Calculation. Percent insoluble matter = $\frac{\text{Weight of residue}}{\text{Weight of sample}} \times 100$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One quart or two quart (1/2 gallon) quantity as specified (see 6.2) of type I cleaning compound shall be filled in a plastic bottle of appropriate size. The bottle shall be fitted with plastic cap possessing a pouring spout with a closure feature. The plastic bottle and cap shall be made of material commonly used by the industry and compatible with the product.

5.1.1.2 Twenty ounces of type II (granular or globular) cleaning compound shall be filled in a round metal can conforming to PPP-C-96, type V, class 2. In lieu of the metal friction plug a tight-fitting polyethylene plug of the friction type shall be used. Exterior protective coating shall conform to plan B.

5.1.2 Level B.

5.1.2.1 One quart or two quarts (1/2 gallon) cleaning compound shall be packaged as described in 5.1.1.1.

5.1.2.2 Twenty ounces of type II cleaning compound shall be filled in one of the containers specified in 5.1.2.2.1, 5.1.2.2.2 or 5.1.2.2.3 at the option of the supplier. One end of each container shall have a metal or plastic closure for a dispenser opening of not less than 3/4 inch in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

5.1.2.2.1 A round metal can shall conform to PPP-C-96, type V, with a plan A exterior non-metallic coating. Alternatively, a can with pour spout, completely lithographed, varnished or lacquered shall be furnished.

5.1.2.2.2 A fiber can shall be a fiber-foil composite can with a minimum wall thickness of 0.028 inch. The foil may be either the inner or outer layer.

5.1.2.2.3 A plastic container shall be a commercial type blow-molded plastic container.

5.1.3 Level C. Cleaning compound shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Twenty-four 1-quart bottles or twelve 2-quart (1/2 gallon) bottles of type I cleaning compound, packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636, style CSSC, class weather resistant. The box shall be fitted with partitions of the design normally used except that the partitions shall be 3/8 to 1/2 inch higher than the bottles.

5.2.1.2 The type II cleaning compound, packaged as specified in 5.1 shall be packed in accordance with the level A packing requirements in the appendix of PPP-C-96.

5.2.2 Level B.

5.2.2.1 Twenty-four 1-quart bottles or twelve 2-quart (1/2 gallon) bottles of type I cleaning compound, packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636, style CSSC or RSC, class domestic. The box shall be fitted with pads and partitions of the design normally used except that the partitions, dividers, or cells as applicable, shall be 3/8 to 1/2 inch higher than the bottles.

5.2.2.2 Twenty-four twenty ounce containers of type II cleaning compound, packaged as specified in 5.1 shall be packed in accordance with the level B packing requirements in the appendix of PPP-C-96, table IIa as specified for type V and VIII - except friction plug cans for one-quart.

5.2.3 Level C. Cleaning compound in quantities as specified (see 6.2) packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall be in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Labeling and marking. The labeling and marking shall comply with the Federal Hazardous Substances Act and the Federal Insecticide, Fungicide, and Rodenticide Act as applicable (see 3.4). In addition to any special markings required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The toilet bowl cleaner covered by this specification is intended to be used in vitreous china surfaces for the removal of stains and lime deposits.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Size and level of packaging and packing required (see 5.1 and 5.2).
(d) Quantity required (see 5.1).
(e) Marking document required (see 5.3).

MILITARY INTEREST:

User activities:

Army - GL

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
GSA-PBO
HEW-FDA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.